SMACT v4: Enhanced Screening and Prediction Tools for Computational Materials Design

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Summary

SMACT (Semiconducting Materials by Analogy and Chemical Theory) is a Python library for the high-throughput screening and discovery of inorganic materials. The name reflects foundational analogy-based approaches to semiconductor prediction (Goodman, 1958; Pamplin, 1964). Originally published as a JOSS paper in 2019 (Davies et al., 2019), the package provided chemical filters based on charge neutrality, electronegativity, and elemental data to rapidly identify chemically plausible compositions (Davies et al., 2016). Since then, SMACT has undergone substantial development from v2.1 through v4.0. The library now includes modules for crystal structure prediction via data-mined ionic substitution (Hautier et al., 2011), dopant prediction, composition-to-property prediction using deep learning models (Goodall & Lee, 2020), and tools for mapping inorganic crystal chemical space (Park, Onwuli, et al., 2025). These additions transform SMACT from a compositional screening tool into a comprehensive platform for low-cost computational materials design.

Statement of Need

The search space for new inorganic materials is immense, exceeding 4×10^{12} quaternary element combinations (Davies et al., 2016), and continues to grow as multi-component systems attract interest. Researchers need tools that can rapidly filter this space using established chemical principles before committing to expensive first-principles calculations or synthesis. The original SMACT addressed this need through charge neutrality and electronegativity screening. However, the field has since advanced: researchers now require not only compositional screening but also structure prediction, property estimation, and integration with modern machine learning workflows.

SMACT v4 meets these needs by providing a unified Python library where a researcher can (i) enumerate and filter compositions, (ii) predict likely crystal structures from data-mined substitution probabilities, (iii) estimate material properties directly from stoichiometry without requiring structural information, and (iv) identify candidate dopants for known host materials. The four stages share a common representation of compositions and oxidation-state-resolved species, so screening output can flow directly into structure, property or dopant prediction. Only structure prediction requires structural input — the parent structures it substitutes into — so a campaign can narrow the chemical space before incurring any structural cost. The library is designed for desktop-scale computation, with data caching and multiprocessing support that enables screening of millions of compositions in minutes.

State of the Field

Several established libraries serve the computational materials science community. `Pymatgen` (Ong et al., 2013) provides comprehensive tools for structure manipulation and thermodynamic analysis. The Atomic Simulation Environment (`ASE`) (Larsen et al., 2017) focuses on atomistic simulations. `Matminer` (Ward et al., 2018) specialises in featurisation and machine learning pipelines for materials data. These tools primarily operate on known structures or experimental data.

SMACT occupies a distinct niche: it works at the composition level, upstream of structure-based tools, to define which regions of chemical space merit further investigation. This might overlap with other tools where they also work on the composition of materials studied. For example, `Pymatgen` implements the same data-mined ionic-substitution model (Hautier et al., 2011), with `SubstitutionPredictor` and `Substitutor` already working at the composition level like SMACT; dopant prediction, however, is provided only by separate functions (`get_dopants_from_substitution_probabilities`, `get_dopants_from_shannon_radii`) that require a structure already decorated with oxidation states. SMACT's native `Doper` class, developed independently, closes that gap by combining the same substitution statistics with oxidation-state-based dopant classification directly at the composition level, so a candidate host never needs a resolved structure before dopants can be suggested for it. Its ranking is further extended by a selectivity term weighing a dopant's affinity for one host site against the others, and by an optional swap to a learned species-embedding similarity score. Nonetheless, the popularity of SMACT comes largely from native validity prediction through charge neutrality, electronegativity rule and data-driven oxidation state filtering which makes it a recognised option for evaluating AI generated materials (Merchant et al., 2023; Zeni et al., 2025). Composition-only bandgap prediction, provided through a pretrained ROOST model with uncertainty estimates, has no equivalent in other libraries and allows candidates to be ordered before any structure is proposed. SMACT brings the full workflow — composition enumeration, validity screening, structure prediction, property prediction and dopant prediction — into one package built on the same `Element`/`Species` classes, rather than requiring a researcher to assemble it from separate tools.

Software Design

SMACT v4 retains the `Element` and `Species` classes at its core, providing access to tabulated data for all elements in arbitrary oxidation states and coordination environments. The following major modules have been added since the original publication.

Structure prediction. Implements data-mined ionic substitution methods (Hautier et al., 2011) to predict likely crystal structures for new compositions by analogy with known parent structures, using substitution probabilities derived from statistical analysis of experimental databases.

Dopant prediction. Enables high-throughput identification of p-type and n-type dopants for a given host composition by combining oxidation state filters and species embedding similarities (Antunes et al., 2022; Onwuli et al., 2024).

Property prediction. Provides composition-to-property prediction using a pre-trained ROOST model (Goodall & Lee, 2020), allowing users to predict band gaps directly from chemical formulae with uncertainty estimates.

Oxidation states and metallicity. Implements a probabilistic model for predicting the likelihood of metal species coexisting in compounds based on anion-dependent statistics (Davies et al., 2018), with updated default data (ICSD24). A metallicity scoring module distinguishes ionic compounds from intermetallic phases.

Crystal space utilities. Tools for systematically mapping inorganic composition space (Park, Onwuli, et al., 2025), including data acquisition, composition generation, and dimensionality-reduced visualisation of compositional embeddings.

Screening enhancements. The core screening functions now support mixed-valence compositions, switchable oxidation state datasets, and consensus-based filtering.

Research Impact Statement

SMACT has been employed in major materials discovery campaigns, including GNoME from Google DeepMind (Merchant et al., 2023) and MatterGen from Microsoft Research (Zeni et al., 2025). Park et al. (Park, Onwuli, et al., 2025) used SMACT's screening and crystal space tools to map the landscape of inorganic crystal chemistry. Onwuli et al. (Onwuli et al., 2024) employed SMACT's species representations for materials informatics, developing ionic embeddings that capture oxidation-state-dependent chemical behaviour. Park et al. (Park, Mastej, et al., 2025) used SMACT in a workflow for closing the synthesis gap in computational materials design. Nduma et al. (Nduma et al., 2025) integrated SMACT into the Crystalyse multi-tool agent for autonomous materials design.

AI Usage Disclosure

Generative AI tools (Claude, Anthropic) were used to assist with code modernisation tasks during the v4.0.0 development cycle. All AI-generated code was reviewed and validated by the authors.

Author Contributions

KOM is the current maintainer who contributed consensus-based oxidation state filtering for `smact_validity` and `smact_filter`, and led the v4.0.0 release including mixed-valence support, a deep code audit, full type annotations, security hardening, test migration to pytest with coverage raised to over 85%, and CI consolidation. AO was the primary developer from v2.2 to v3.1, contributing the dopant prediction module, oxidation states improvements, crystal space utilities, and extensive maintenance. AM designed and implemented the structure prediction subpackage and benchmarking framework. DWD is the original lead developer and contributed to CI infrastructure and code quality tooling. RN implemented the property prediction subpackage, metallicity module, and screening performance optimisations. KTB is an original author who continued contributing to the codebase. JL contributed the initial dopant prediction implementation and charge state comparison fixes. MN updated and fixed all example and tutorial notebooks to work correctly with the current SMACT library. PD contributed to lattice parameter calculations. AMG contributed to the implementation of mixed-valence support during the Alchemy's SMACT hackathon. HP contributed the crystal space visualisation tools and element data corrections. REAG fixed a screening bug related to electronegativity handling. TL contributed to code reorganisation and utility refactoring. AW supervised the project and contributed to the codebase.

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